



DFS CORPORATE PORTAL

FUNCTIONAL SPECIFICATION DOCUMENT

Functional Specification Document

Actors, screens, APIs, data, workflows, and integrations for Bank / EMI IT and delivery review.

DOCUMENT ID

DFS-CP-FSD-001

VERSION

1.0 — Draft for Client Review

AUDIENCE

Bank / EMI Client (IT · Ops · Delivery)

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PRODUCT

DFS Corporate Portal

CLASSIFICATION

Client Confidential

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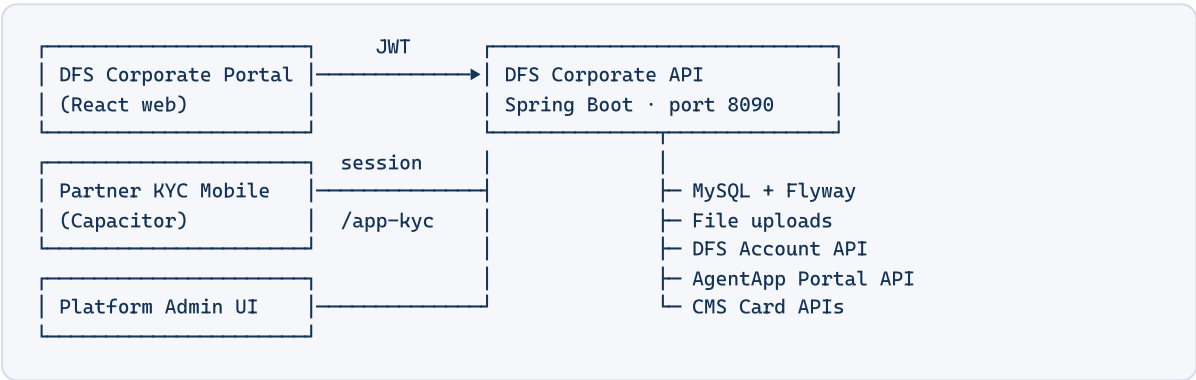
1. Purpose & scope

This FSD specifies how DFS Corporate Portal behaves: actors, roles, screens, APIs, data, workflows, integrations, and non-functional characteristics.

In scope: Web portal, platform admin, partner mobile KYC APIs, franchise flows, portal approvals, DFS/CMS integrations.

Out of scope: Source listing, internal class design, vendor commercial contracts (TBD).

2. System context



Layer	Technology
Web portal	Vite, React 19, TypeScript, React Router
Mobile KYC	Vite + React + Capacitor (com.dfscorporate.kyc)
API	Spring Boot 3.3, Java 21, Spring Security, JWT, JPA
Database	MySQL 8 (default), Flyway
Files	Local disk upload directory (Docker volume)

3. Actors, roles & permissions

3.1 Account roles

Role	Description	Access
PLATFORM_ADMIN	Platform back-office	/admin, /api/admin/**
PARTY_USER	Corporate party user	Portal APIs scoped to party

3.2 Portal workflow roles

Role	Capability
PARTY_ADMIN	Manage portal users; elevated within party
MAKER	Create approval requests
CHECKER	First review
APPROVER	Second review (skippable if checker also Approver)
RELEASER	Final release

Sequence: MAKER → CHECKER → APPROVER → RELEASER → DONE

Types: GENERIC, PAYMENT, BULK_PAYMENT, COMMISSION_CHANGE

TBD: Payment execution rails — approval types exist; end-to-end payment settlement is not specified as live.

3.3 Party types & UI gating

Type / area	Rule
MERCHANT	Self-signup
SUB_MERCHANT	Franchise invite from parent
Finance screens	Party ACTIVE (Merchant)
Network / ops	MERCHANT + ACTIVE
Admin	PLATFORM_ADMIN

4. Functional modules

4.1 Public & authentication

Method	Path	Notes
POST	/api/auth/signup	Merchant signup
POST	/api/auth/verify-otp	Email OTP
POST	/api/auth/login	JWT (~24h default)
GET	/api/party-types, entity-types, id-document-types	Reference LOVs
GET	/api/party-types/{type}/documents	Required docs matrix
GET	/api/public/franchise-invite/{token}	Validate invite

4.2 Entity onboarding wizard

Route: /onboarding · **Steps:** Entity → Partners (conditional) → Documents → Review / Submit

Capability	Detail
Entity CDD	Legal name, type, incorporation, NTN/tax, FATCA/CRS, addresses, purpose, T&Cs
Geo evidence	IP + browser geolocation when permitted
Partners	Partnership/LLP roster
Documents	Multipart Annex-C uploads
Submit	Tracking ID; 5 working-day due; status progression
Statuses	DRAFT → SUBMITTED / PENDING_APPROVAL → ACTIVE REJECTED (SUSPENDED supported)
Draft window	30 days

Entity types: SOLE_PROPRIETORSHIP, SMALL_BUSINESS, PARTNERSHIP, LLP

Entity	Typical mandatory evidence
Sole / Small	Owner ID front/back/photo + alternative business evidence
Partnership	Deed, authority, registration cert (unless unregistered)
LLP	Deed, SECP certificate, authority
Sub-merchant	Parent authority + authorised ID pack

4.3 Partner mobile KYC

Deep link: dfscorporate://kyc?token=... · API prefix: /api/public/app-kyc

Step	Action	Result
1	Login / invite	Session
2	OTP send / verify	Verified; LOVs; video script
3	Change password	Temp PIN → password
4	Video challenge / verification	Video evidence
5	Submit multipart media	KYC_COMPLETED

Media: CNIC_FRONT, CNIC_BACK, SELFIE, FINGER_L1–L4, FINGER_R1–R4 · CNIC without dashes (13 digits).

Portal = business documents. Mobile app = personal KYC. Secondary web partner KYC (/partner-kyc/:token) remains available.

4.4 Platform administration

Route: /admin · **Role:** PLATFORM_ADMIN

Function	Description
Queues	Pending / all parties with Tracking ID & TAT
Party detail	Profile, persons, docs, IP/geo, risk, screening, identity
Sanctions / identity	Manual CLEAR / HIT; manual / waived identity
Documents	Approve / reject / download
Decision	Approve (ACTIVE + credentials) / Reject (reason)
Provision	Retry DFS corporate onboarding
Discrepancy	Notes to applicant

4.5 Post-activation Merchant portal

Route	Function	Integration
/dashboard	Status, KYC, provision, KPIs	Composite
/balance	Agent balance	Portal API
/statement	Mini-statement	Portal API
/transactions	Franchise / commission view	Franchise services
/cards	Card search / inquiry / status	CMS
/invites	Franchise invites	Franchise APIs
/franchises	Network, commission confirm, child views	Franchise + Portal
/approvals	Create & decide requests	Workflow APIs
/portal-users	Users & portal roles	Portal user APIs
/profile	Party profile	Onboarding me

4.6 Franchise / commission

Parent sets proposed commission % → child onboards as SUB_MERCHANT → parent locks commission plan. Ledger amount tables removed in later schema; **% plans** remain. Settlement engine: **TBD** .

5. End-to-end workflows

5.1 Merchant onboarding → provision

MERCHANT signup → OTP → /onboarding (Entity → Partners → Docs)
→ Submit (Tracking ID, decision_due_at)
→ Partner app invites → each App KYC → KYC_COMPLETED
→ PENDING_APPROVAL → Admin review → Approve
→ ACTIVE + credentials → DFS corporate onboarding (if enabled)
→ accountProvisionStatus SUCCESS | FAILED | PENDING

5.2 Sub-merchant

Parent /invites → token → /franchise-onboard → SUB_MERCHANT signup
→ OTP → onboarding (+ PARENT_AUTH) → KYC → Admin → ACTIVE → provision
→ Parent confirms commission plan

5.3 Approval workflow

PARTY_ADMIN assigns roles → MAKER creates approval_request
→ CHECKER → APPROVER (may skip) → RELEASER → APPROVED/REJECTED
→ Actions stored on approval_actions

6. Data specification (logical)

Entity	Purpose
parties	Corporate applicant / activated customer
accounts	Login users (multi-user per party)
account_portal_roles	Maker-checker roles
otp_codes	OTP challenges
required_documents / party_documents	Matrix + uploads
associated_persons	Partners / related persons
partner_invites	Legacy web partner KYC tokens
partner_app_users	Mobile KYC users & session
video_challenges	Read-aloud video challenges
franchise_invites / franchise_commission_plans	Invites + locked % plans
approval_requests / approval_actions	Workflow
brands	Soft multi-tenant brand

7. Integrations

Integration	Flag / key	Functions
DFS Account API	DFS_ACCOUNT_API_ENABLED	corporateonboarding, LOVs, segments
AgentApp Portal API	DFS_PORTAL_API_ENABLED + portal key	Balance, mini-statement, change MPIN
CMS Card APIs	DFS_CMS_API_ENABLED	Search / status / inquiry / LOVs
Email	MAIL_ENABLED	OTP, invites, credentials

TBD: Environment URLs and credentials per client environment.

8. Non-functional specification

Area	Specification
Auth	Stateless JWT; BCrypt for stored passwords
App KYC session	Separate session tokens on partner app users
Uploads	Configurable large multipart limits
Health	/actuator/health
Sensitive handling	Partner password plain until DFS provision (integration constraint) — harden in production security review TBD
Multi-tenancy	brand_id soft tenancy — hard isolation TBD
Retention / encryption at rest	TBD
Availability SLA	TBD
Accessibility / i18n	Not formally certified — TBD

9. Screen inventory (web)

Route	Screen	Auth
/	Home	Public
/getting-started	Getting started	Public
/signup	Merchant signup	Public
/franchise-onboard	Franchise signup	Public (token)
/verify-otp	OTP verify	Public
/login	Login	Public
/onboarding	KYC wizard	Party user
/partner-kyc/:token	Web partner KYC	Token
/admin	Back-office	Platform admin
/dashboard	Dashboard	Authenticated
/balance,/statement,/transactions,/cards	Finance	ACTIVE
/invites,/franchises,/approvals,/portal-users	Network / ops	MERCHANT ACTIVE
/profile	Profile	Authenticated

10. Error & status handling

Situation	Expected behaviour
Invalid OTP	Reject; allow resend per policy
Incomplete Annex-C pack	Block submit with validation
Partners incomplete	Do not enter PENDING_APPROVAL until app KYCs complete
Admin reject	REJECTED; reason to applicant
Provision failure	May remain ACTIVE; provision FAILED; admin retry
Portal/CMS disabled	Controlled unavailable — no silent success

11. Traceability (BRD → FSD)

BRD ID	FSD coverage
BR-ONB-*	\$4.1–4.3, \$5.1–5.2
BR-CMP-*	\$4.4
BR-PRT-*	\$4.5–4.6, \$5.3
BR-NFR-*	\$8

12. Open functional items

1. Live NADRA BV/Verisys replace manual identity path — TBD
2. Automated sanctions vendor replace manual CLEAR — TBD
3. Payment / bulk-payment approval execution — TBD
4. Formal retention, purge, encryption standards — TBD
5. Hard multi-brand isolation if required — TBD
6. Production notification channel policy — TBD
7. UAT scripts & regulatory evidence pack — TBD